

2. $\mathcal{X} = \mathbb{R}^2$

let $x = (x_1, x_2)$, $y = (y_1, y_2)$ and let $\phi(x) = (x_1^2, \sqrt{2}x_1y_1, x_2^2)$.

1. Verify that $\phi(x)^T \phi(y) = (x^T y)^2$.

$$(x_1y_1 + x_2y_2)^2 = (x_1y_1)^2 + (x_2y_2)^2 + 2x_1x_2y_1y_2$$

$$\begin{aligned} \phi(x)^T \phi(y) &= (x_1^2, \sqrt{2}x_1x_2, x_2^2) \cdot (y_1^2, \sqrt{2}y_1y_2, y_2^2) = \\ &= (x_1y_1)^2 + 2x_1x_2y_1y_2 + (x_2y_2)^2 = (x_1y_1 + x_2y_2)^2 \end{aligned}$$

2. (a) Find $\phi(x) : \mathbb{R}^2 \mapsto \mathbb{R}^6$ s.t. for any (x, y) , $\phi(x)^T \phi(y) = (x^T y + 1)^2$.

Since $(x^T y + 1)^2 = x_1^2 y_1^2 + x_2^2 y_2^2 + 2x_1x_2y_1y_2 + 2x_1y_1 + 2x_2y_2 + 1$, take $\phi(x) = (x_1^2, x_2^2, \sqrt{2}x_1x_2, \sqrt{2}x_1, \sqrt{2}x_2, 1)$

(b) Find $\phi(x) : \mathbb{R}^2 \mapsto \mathbb{R}^9$

Take $\phi(x) = (x_1^2, x_2^2, \sqrt{2}x_1x_2, \sqrt{2}x_1, 1/\sqrt{5}, 1/\sqrt{5}, 1/\sqrt{5}, 1/\sqrt{5}, 1/\sqrt{5})$. Verify that

$K_l(x_i, x_j) = (1 + x^T y)^d$ is pos. definite for $d = 1, 2, \dots$

(a) $K_d(x_i, x_j)$ is symmetric since $K_d(x_i, x_j) = K_d(x_j, x_i)$ by the sym property of the inner product

(b) $K_1(x, y) := K$ is positive definite since for any $v \in \mathbb{R}^n$ we have

$$\begin{aligned} v^T K v &= \sum_{ij}^n v_i v_j k(x_i, x_j) = \sum_{ij} v_i v_j (1 + x_i^T x_j) \\ &= \sum_i \sum_j v_i v_j + \sum_i \sum_j v_i v_j x_i^T x_j \\ &= \sum_i v_i \sum_j v_j + \left(\sum_i v_i x_i^T \right) \left(\sum_j v_j x_i \right) \\ &= \left(\sum_k v_k \right)^2 + \left(\sum_i v_i x_i \right)^T \left(\sum_j v_j x_i \right) \geq 0 \end{aligned}$$

If $w := \sum_i v_i x_i$ then the sum on the right is the norm of w and so both terms are greater or equal to zero. (b) continued

Assume $k_{d-1}(x, y) = (1 + x^\top y)^{d-1}$ is pos. definite.

Then $k_d(x, y) = (1 + x^\top y)^d = (1 + x^\top y)^{d-1} (1 + x^\top y)$ is also pos. definite by (c) below. Since $(1 + x^\top y)$ is pos definite, by induction $K_d(x, y) = (1 + x^\top y)^d$ is also.

(c) If $k_1(x, y)$ and $k_2(x, y)$ are pos. definite that so is their product $k = k_1 \cdot k_2$
Proof.

Define the following matrices:

C is the matrix with elements, $c_{ij} = k_1(x_i, x_j)$,

D is the matrix with elements, $d_{ij} = k_2(x_i, x_j)$

E as the matrix formed by the elements, $e_{ij} = c_{ij}d_{ij}$,

E is clearly symmetric since k_1 and k , are both symmetric

Pick and $u \in \mathbb{R}^2$. Since C and D are pos. definite they can be decomposed as

$$C = A^\top A \text{ and } D = B^\top B$$

for some matrices A, B .

Now write

$$A = (a_1 \ a_2 \ \cdots \ a_n), B = (b_1 \ b_2 \ \cdots \ b_n)$$

where a_i and b_i are the columns of A, B respectively. Therefore,

$$c_{ij} = a_i^\top a_j = \sum_k a_{ik}a_{jk}, \quad d_{ij} = b_i^\top b_j = \sum_l b_{il}b_{jl}$$

Finally,

$$\begin{aligned} u^\top E u &= \sum_{ij} u_i u_j c_{ij} d_{ij} = \sum_{ij} u_i u_j \left(\sum_k a_{ik} a_{jk} \right) \left(\sum_l b_{il} b_{jl} \right) \\ &= \sum_{kl} \sum_{ij} u_i u_j a_{ik} a_{jk} b_{il} b_{jl} \\ &= \sum_{kl} \left(\sum_i u_i a_{ik} b_{il} \right) \left(\sum_j u_j a_{jk} b_{jl} \right) = \sum_{kl} f_{kl}^2 \geq 0 \end{aligned}$$

(since Dummy indices don't care what you call them)

(d) Let $x = (x_1, x_2)$ and

$$\phi(x) = \begin{cases} 1, d = 0 \\ (\phi_{d-1}(x), x_1\phi_{d-1}(x), x_2\phi_{d-1}(x))^\top, d = 1, \dots \end{cases} \quad (1)$$

For $d = 1$, the kernel $K_d(x, y) = \phi(x)_d^\top \phi(y)_d = (1 + x^\top y)$ is well defined, see (a) above. For $d > 1$ the feature map can be written as $\phi_d(x) = (1, x_1, x_2)\phi_{d-1}(x)$. Assume that $\phi(x)_d^\top \phi(y)_d$ is well defined. Then since $\phi(x)_{d+1}^\top \phi(y)_{d+1} (1 + x^\top y) \phi(x)_d^\top \phi(y)_d$ by (c) above and induction $K_d(x, y)$ is well-defined for all d .

4. (a) Let $K(x, y)$ be pos. definite. Then for any $u \in \mathbb{R}^2$
 $u^\top k(x, y) u \geq 0 \Rightarrow$ for $k'(x, y) = -k(x, y) \Rightarrow u^\top k'(x, y) u \leq 0$

(b) Find a $\phi(x) : \mathbb{R}^2 \rightarrow H$ where for any x, y

$$\phi(x)^\top \phi(x) = x^\top y - 1.$$

Proof: None exist Since this must hold for $x = (0, 0)$ and $y = (0, 0)$ which would imply that

$$\phi(x)^\top \phi(x) = \|\phi(x)\|^2 = -1$$

This is impossible since $\|\cdot\| \geq 0$